Financial vulnerability and social reward

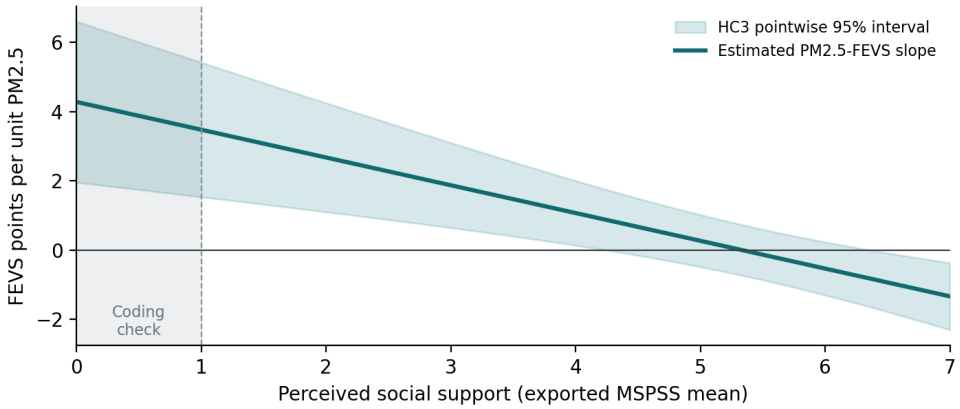
RF1-SRA / Katie Social Doors | Adjusted results brief | November 4, 2026

PRELIMINARY: financial-score verification and MSPSS coding checks remain open. N=97 and settled exclusions are unchanged.

Working synthesis: environmental context and perceived support are associated jointly with financial vulnerability; separate neural analyses relate FEVS to socially differentiated reward responses. This is an exploratory set of contextual and neural correlates, **not a demonstrated causal or mediating pathway**.

1. Support Moderates The PM2.5-FEVS Association

Interaction b = **-0.802**; HC3 95% CI **[-1.208, -0.396]**; two-sided p = **0.000168**; Holm p = **0.000336** across the two primary behavioral tests.



The plotted quantity is the estimated PM2.5-FEVS slope, not the FEVS score itself. The association is more positive at lower support and weakens as support increases. Bands are pointwise, not simultaneous. The shaded 0-1 region flags values below the standard MSPSS mean-score range.

Matched ADI/Ecog Adjustment Checks

Added covariates	Interaction b	HC3 95% CI	HC3 p
baseline	-0.802	[-1.208, -0.396]	0.000168
adi	-0.802	[-1.217, -0.388]	0.000224
ecog	-0.714	[-1.134, -0.294]	0.001093
adi+ecog	-0.711	[-1.137, -0.285]	0.001313

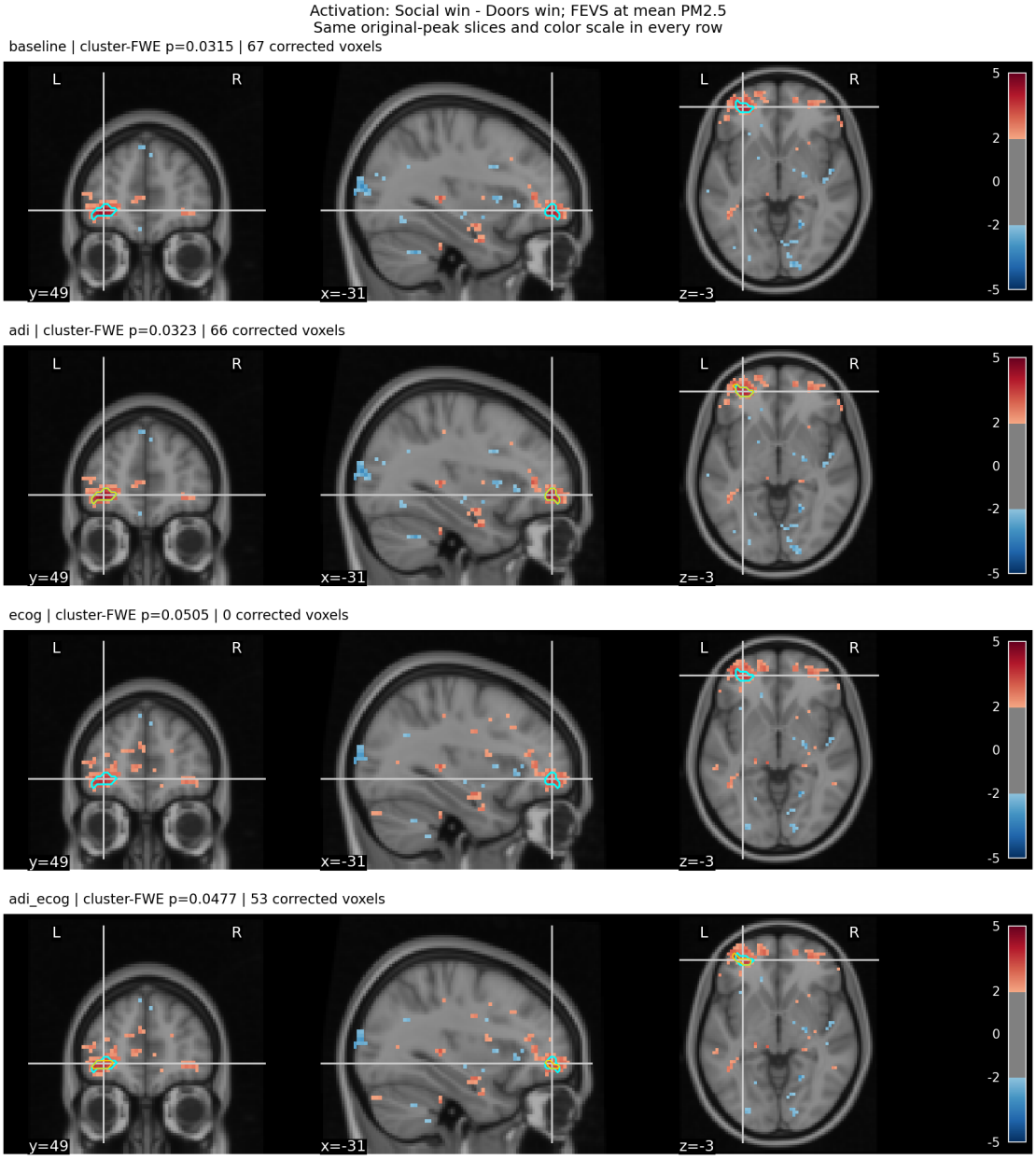
All fits retain N=97 and PM2.5 + MSPSS + their interaction + age + gender. Added covariates are additive, not new interactions. Sensitivity p values above are unadjusted across these related checks. Every single-case omission retains a negative interaction in the baseline and all adjusted fits. This does not resolve sensitivity to multiple questionable scores together.

Scoring remains provisional: five exported MSPSS values fall below the standard 1-7 mean range, including one zero. They remain as supplied; no imputation or exclusions. FEVS/OAFEM instrument versions, items and timepoints also need sign-off. None of the other seven behavioral interactions survives its within-family Holm correction.

Behavioral model family | All matched behavioral adjustments

2. Activation: Stable Location, Qualified Evidence

ACTIVATION | Social Doors win - nonsocial Doors win



FEVS coefficient at mean PM2.5 in the FEVS + PM2.5 + FEVS x PM2.5 model. Original, +ADI, +Ecog, +both; same slices/color scale. Cyan: original corrected cluster; gold: adjusted corrected cluster. Background |t|>=2 shading is descriptive, **not** corrected significance.

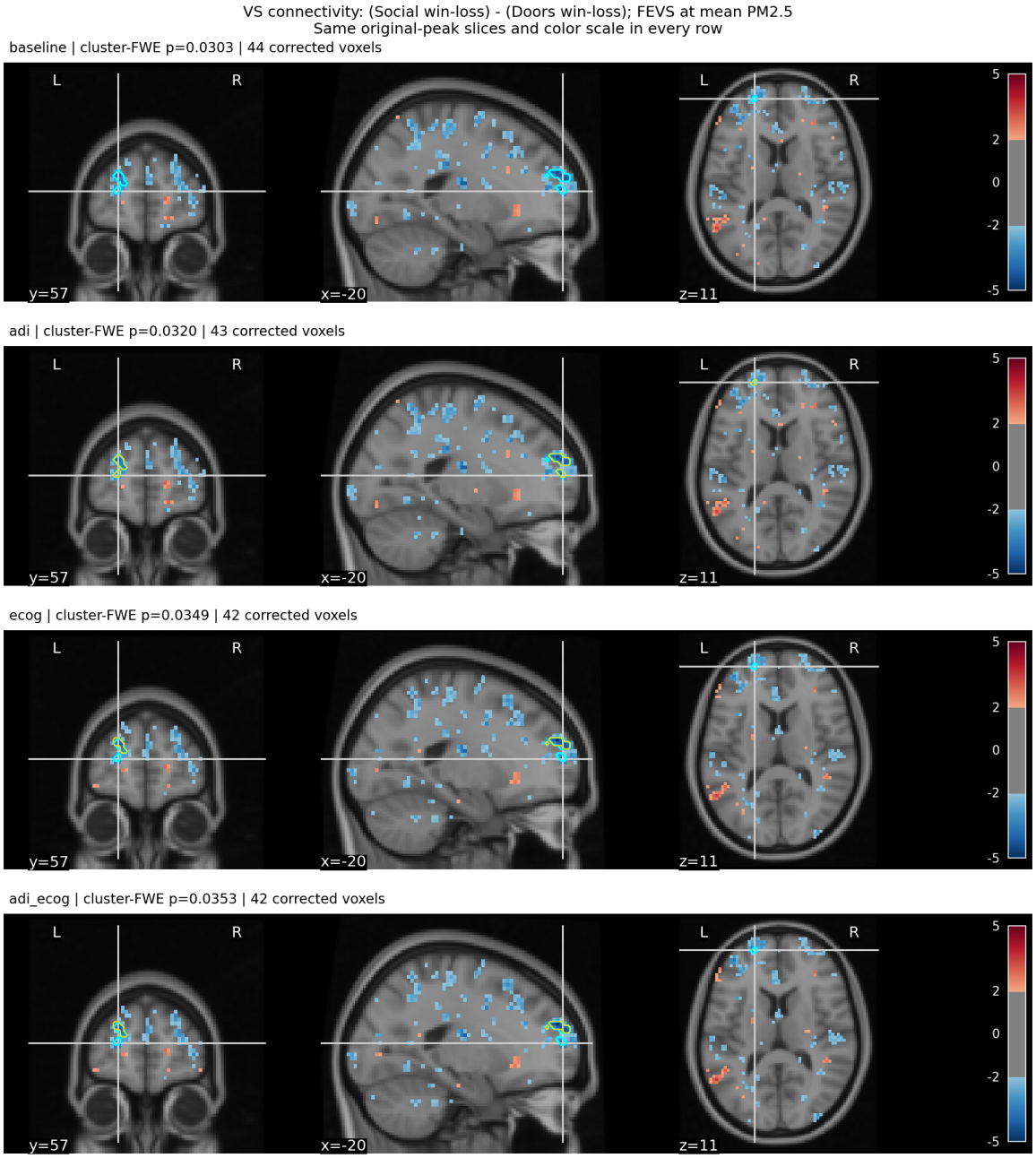
ADI alone leaves the location nearly identical (Dice=.977). Adding both retains 53 voxels, 43 overlapping the original 67; centroid shift=1.1 mm. The 14.8-mm peak shift stays inside the original cluster. Ecog alone narrowly misses cluster correction (p=.0505); both p=.0477.

Activation caveat: standalone FEVS activation also fails correction with Ecog (p=.0666) or both (p=.0709). The adjusted cluster is approximately 64% white-matter-labeled in a standard atlas, not native tissue segmentation. Describe left ventral/anterior frontal localization, **not clean cortical vmPFC/OFC activation**. This is not a winning-related FEVS x PM2.5 interaction.

All activation and connectivity adjustments, including nulls

3. VS Connectivity: More Consistent Across Adjustments

VS PPI | [Social win - social loss] - [nonsocial win - nonsocial loss]



Negative FEVS coefficient at mean PM2.5, not the FEVS x PM2.5 interaction. Rows and contour meanings match page 2. Higher FEVS relates to a lower social-by-valence VS-connectivity contrast, not globally weaker connectivity or winning-specific coupling.

Baseline p=.0303; +ADI p=.0320; +Ecog p=.0349; +both p=.0353. All 43 ADI-adjusted voxels lie in the original 44-voxel cluster. Ecog/both retain 42 voxels, with 32 overlapping the original (Dice=.744). Centroid shift=1.3 mm with both. Every adjusted peak remains inside the original cluster despite an approximately 15-mm peak shift.

The adjusted target remains predominantly left frontal-pole cortex (approximately 81% cortex-labeled; adjusted peak [-20.8, 54.9, 26.0] mm). This frontal target is **not the VS seed**. Both neural findings pass only at CDT 3.1, not 2.6; per-map correction is not correction across the exploratory family. Verified maps, locations and overlap

A Coherent, Provisional FEVS-Centered Story

CONTEXT + FINANCIAL VULNERABILITY + SOCIAL REWARD | Complementary associations, not mediation

The strongest working synthesis is that the PM2.5-FEVS association varies with perceived social support, while FEVS also relates to socially differentiated neural responses. The connectivity association is more consistent across ADI/Ecog adjustment than activation. These are linked by the same financial phenotype, **not a demonstrated biological pathway**.

Variable	Role in this brief	Descriptive r with FEVS
FEVS	Lead financial measure: behavioral outcome; neural predictor. Scoring provisional.	--
PM2.5	Environmental exposure in the behavioral support interaction; retained in conditional neural fits.	.11
MSPSS	Behavioral moderator. No matching corrected neural support interaction.	-.32
ADI	Environmental context and additive sensitivity covariate; not a second headline association here.	.09
Ecog	Everyday-cognition measure; adjustment tests shared variation, not proof of longitudinal decline.	.39
OAFEM	Secondary financial measure; no equivalent behavioral moderation result survives family correction.	.33

Correlations are unadjusted Pearson descriptions of the same provisional exports, not model selection or independent evidence. The weak marginal PM2.5/ADI correlations do not invalidate a conditional association. Shared Ecog/FEVS variation does not prove confounding.

Keep The Neural Outcomes Distinct

Social winning: social win - nonsocial win (activation finding). **Domain-by-valence:** [social win-loss] - [nonsocial win-loss] (VS-connectivity finding). These are not interchangeable evidence for one replicated social-reward effect. There is **no matching corrected neural PM2.5 x support interaction**, no FEVS social-winning VS-connectivity finding, and no domain-by-valence activation finding.

Scope And Reporting Boundaries

Exploratory, not preregistered. Frozen N=97, paired ses-01, age 55+; completed no-TEDANA sm-5 FEAT inputs and original VS seed. Neural models retain age, gender, scanner era and signed FD/tSNR differences. Age is a covariate, not a lifespan or longitudinal effect here. Cluster-extent randomise: 10,000 permutations; Z-equivalent CDTs 2.6/3.1; cluster-FWE per map only. Sensitivity checks are related models, not independent replications. No new three-way interactions, exclusions or inferential fits were added for this brief.

Closeout Before Final Interpretation

Verify MSPSS item coding and all 97 scores, including the five values below 1; confirm FEVS/OAFEM versions, coding, timepoints and missing-item rules. Do not replace zeros, impute or exclude people to resolve coding. Preserve original exports and all 97 participants; version any verified corrections before assessing their implications. Computation, adjusted-map review and publication are complete.

Complete neural inventory, nulls and secondary findings | All matched neural adjustments | Descriptive correlations